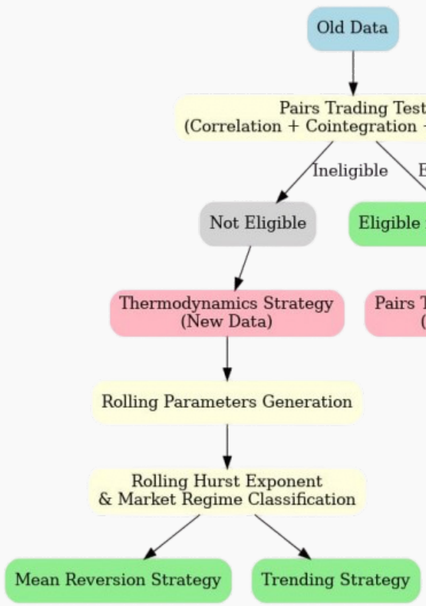
1. PROBLEM STATEMENT

The PS is about designing a trading bot that uses limit order book and trade log data for seven commodities, respects position/liquidity constraints, and generates strategies to Maximize PnL while uncovering hidden patterns in market microstructure

2. STRATEGY FRAMEWORK



3. PAIRS TRADING STRATEGY

3.1 Introduction

Pairs trading is a market-neutral strategy where we look at the relationship between two stocks instead of the whole market. The idea is simple: if two stocks usually move together but drift apart, we can short the one that goes too high and buy the one that goes too low, waiting for them to come back together. In this project,We started with 7 assets that were given in the problem statement. The task was to find the best two assets that can be traded as a pair. For this, I used three tests: **Correlation**, **Cointegration**, and the **Augmented Dickey-Fuller (ADF)**.

3.2

PAIR IDENTIFICATION		
Correlation	**Cointegration**	**ADS**
Pearson correlation on log returns to find strongly related assets.	Engle–Granger test to check for long-run equilibrium (reject H_0 if p-value < 0.05	Augmented Dickey–Fuller test on residuals to ensure the spread is stationary (mean-reverting).

3.3 Z-Score Model

Only the pair that passes all three tests is retained. Thus, although correlation is the starting point, the final decision is made on the basis of both cointegration and stationarity of residuals. All trading thereafter is restricted exclusively to this selected pair.

After the selection step, we model the relationship using the ratio of the two asset prices. Suppose X_t and Y_t are the chosen prices; then the ratio is defined as

$$\boxed{r_t = \frac{X_t}{Y_t}.}$$

where Y is price of asset Y[V-Wap] and X is price of asset X[V-Wap]. We then compute

$$\boxed{Z(t) = \frac{S(t) - \mu_S}{\sigma_S},}$$

with μ_S, σ_S as the rolling mean and standard deviation. Interpretation:

- $Z > 1$: Spread too high $\Rightarrow$ Short Asset 1, Long Asset 2.
- $Z < -1$: Spread too low $\Rightarrow$ Long Asset 1, Short Asset 2.
- $-1 \leq Z \leq 1$: Hold / no trade.

3.4 Signal Generation & Restrictions

Trading signals come directly from Z-score thresholds. We **only trade the identified pair**, ignoring all other assets. Position sizes respect the ± 50 constraint from the PS.

3.5 No Look-Ahead Bias

Our backtester uses only:

- Order book and price data up to time t.
- Trade log data strictly up to $t - 1$.

This guarantees **fair evaluation** and **prevents future leakage**.

3.6 Results (Graphs Placeholder)

- Spread vs time with mean and thresholds.
- Z-score evolution.
- Entry/exit trade markers.

3.7 Conclusion

We identified a robust trading pair using three tests (correlation, cointegration, ADF). A Z-score based framework generated reliable long/short signals. **Our method is statistically sound, avoids look-ahead bias, and obeys all position rules**. This disciplined approach maximizes PnL under Takneek's constraints.

4. THERMODYNAMICS STRATEGY

4.1 Abstract

This section introduces a **thermodynamic perspective** on the Limit Order Book (LOB), where market activity is summarized using two key variables: **Market Temperature** and **Market Entropy**. Together, these measures provide a compact yet powerful framework to classify market regimes, distinguish between order and disorder in trading activity, and guide **systematic decision-making**.

4.2 Market Temperature

Market Temperature (T) quantifies the intensity of price fluctuations, weighted by liquidity at the top levels of the order book. It effectively captures the combined impact of **volatility** and **order concentration**:

$$\boxed{T = \text{Volatility} \times \log\left(1 + V_{\text{top3}}\right)}$$

where V_{top3} is the total order volume aggregated across the top three bid and ask levels.

To distinguish between calm and active markets, a **threshold temperature** T' is estimated using **Parameter Generator (PG)**. The classification rule is:

$$\boxed{T > T' \Rightarrow \textbf{Hot Market (high activity)}, \quad T < T' \Rightarrow \textbf{Cold Market (low activity)}}$$

A hot market typically signals aggressive trading and heightened price risk, while a cold market reflects reduced participation and slower dynamics.

4.3 Market Entropy

Market Entropy (S) is derived from Shannon entropy and measures the degree of **uncertainty or disorder** in order placement. It evaluates how evenly volumes are distributed across different price levels:

$$\boxed{S = -\sum_{i=1}^{6} p_i \log_2(p_i)}, \quad \boxed{p_i = \frac{\text{Volume at level } i}{\text{Total Volume (top 3 bid + ask)}}}$$

Here, higher entropy indicates a more scattered and unpredictable order book, while lower entropy points to concentrated and structured liquidity.

A **threshold entropy** S' separates ordered from disordered states:

$$\boxed{S > S' \Rightarrow \textbf{Disordered Book}, \quad S < S' \Rightarrow \textbf{Ordered Book}}$$

4.4 Market Regimes

By jointly analyzing (T, S), the market can be mapped into **four distinct regimes**:

Temperature (T)	**Entropy (S)**	**Market Regime / Implication**
High ($T > T'$)	High ($S > S'$)	Chaotic and noisy → **Avoid trading**
High ($T > T'$)	Low ($S < S'$)	Strong directional flow → **Trend-following opportunities**
Low ($T < T'$)	High ($S > S'$)	Illiquid and scattered → **Unreliable market, avoid trading**
Low ($T < T'$)	Low ($S < S'$)	Calm and ordered → **Favourable for mean-reversion**

4.5 Conclusion

The thermodynamic backtester simplifies complex order book dynamics into two measurable variables: **Temperature** and **Entropy**. This dual framework provides a structured way to classify markets into chaotic, trend-driven, illiquid, or mean-reverting regimes. By reducing noise and highlighting structure, it creates a more **interpretable and adaptive approach** for trading strategy design.

5. Rolling Parameter Generator

Financial markets are constantly changing, which means that a fixed set of trading rules quickly becomes outdated. To deal with this, the strategy uses a **walk-forward (rolling) optimisation** approach. Here, Parameter Generator (PG) is not run once on the full dataset, but is instead re-applied on moving windows of past data. This makes sure the thresholds for identifying market regimes are always adjusted to the latest conditions.

5.1 Introduction

Choosing appropriate thresholds for regime definition is not straightforward, since many different pairs of values for temperature and entropy are possible. To address this, we employ **Parameter Generator (PG)**, an optimization technique inspired by the collective behavior of flocks of birds or schools of fish.

In PG, a group of candidate solutions, called particles, explores the search space simultaneously. Here, each particle represents a possible pair of thresholds for temperature and entropy. Initially, the swarm is randomly initialized within reasonable bounds, selected from empirical quantiles of the temperature and entropy distributions.

Each particle evaluates its position according to the objective function and keeps track of two references:

- **Personal Best (pBest):** The best threshold pair discovered by the particle itself so far.
- **Global Best (gBest):** The best threshold pair discovered by the entire swarm.

5.2 Thermodynamic Perspective

The method models markets using thermodynamic analogies, defining **temperature** (volatility) and **entropy** (order vs randomness) to classify regimes: Hot Ordered, Hot Disordered, Cold Ordered, and Cold Disordered. Thresholds on these features split the market into regimes, which are then tested for their ability to predict future return directions, with a **deadband** filter to ignore small, noisy moves.

5.3 Interpretation of Components

- **Inertia Term:** $w \cdot v_i^{(t)}$ This keeps a fraction of the previous velocity, allowing particles to maintain momentum and continue exploring the space instead of stopping abruptly.
- **Cognitive Component:** $c_1 \cdot r_1 \cdot (pbest_i - x_i^{(t)})$ This term pulls each particle toward the best threshold values it has personally discovered so far, promoting individual learning and self-correction.
- **Social Component:** $c_2 \cdot r_2 \cdot (gbest - x_i^{(t)})$ This attracts particles toward the globally best solution found by the swarm, encouraging collective learning and convergence.

5.4 Summary

In short:

$$\text{Velocity Update} = \underbrace{w \cdot v}_{\text{Inertia}} + \underbrace{c_1 \cdot r_1 \cdot (pbest - x)}_{\text{Cognitive}} + \underbrace{c_2 \cdot r_2 \cdot (gbest - x)}_{\text{Social}}$$

5.5 Outcome and Significance

The process yields informative, asset-specific thresholds that partition markets into regimes linked to future returns, reflecting transitions between order and disorder under varying volatility. This blends information theory and swarm intelligence within a thermodynamic framework to reveal interpretable market structure.

5.6 Walk-Forward Idea

Rather than fitting parameters once and assuming they stay valid forever, the data is split into a **training window** and a **testing window** that slide forward through time:

- **Training window ($T_{\textbf{train}} = 700$ observations):** Used to learn regime-detection thresholds.
- **Testing window ($T_{\textbf{test}} = 600$ observations):** Used to apply those thresholds and see how the strategy would have performed on unseen data.

Once the test period finishes, both windows move forward. This step-by-step procedure **avoids look-ahead bias** and reflects how trading decisions are actually made in real time.

5.7 What PG Does in Training

In each training window, PG searches for the best thresholds for *temperature* and *entropy*. These thresholds separate the market into regimes such as "trending" or "mean-reverting."

Each candidate solution is a particle that holds a pair (θ_T, θ_S). The swarm of particles explores the search space by combining:

- **Inertia**

- **Cognitive learning**
- **Social learning**

This mix of self-exploration and cooperation helps PG quickly find threshold values that work well on the training data.

5.8 Applying the Learned Thresholds

Once PG has found thresholds, they are tested on the next window of data:

- The market is labelled into regimes (for example, high vs. low entropy periods).
- Thermodynamic risk measures are scaled using only the training window, to keep the test phase realistic.
- A combined strategy based on the Hurst exponent and thermodynamic measures produces buy, sell, or hold signals.

These signals are then added back into the overall time series of trading decisions.

5.9 Rolling Process in Action

After finishing one test window, the process moves forward:

$$t \;\mapsto\; t + T_{\text{test}}.$$

PG is run again on the most recent training data, and the newly discovered thresholds are applied to the next unseen test window. This cycle creates an adaptive strategy that constantly updates itself in response to changing markets.

5.10 Why Rolling PG Matters

Markets are *non-stationary*, meaning their behaviour changes over time. A fixed set of parameters might work for a while but will eventually fail. Rolling PG solves this by:

- **Staying adaptable:** thresholds evolve with shifts in volatility, entropy, and market structure,
- **Being robust:** the walk-forward design reduces the risk of overfitting,
- **Remaining realistic:** decisions are always made using past data, just as they would be in live trading.

6. Hurst Exponent and Market Regime Identification

6.1 The Hurst Exponent

The Hurst exponent (H) provides a statistical measure of persistence or mean reversion in financial time series. By combining H with the concept of market temperature, regimes of trending and mean-reverting behavior can be systematically identified. In financial markets, Hurst exponent serves as a measure of the correlation structure of returns. It is calculated using the rescaled range analysis:

$$\boxed{H = \frac{\log(R/S)}{\log(n)}}$$

where R is the range of cumulative deviations from the mean, S is the standard deviation, and n is the time window size. Values of H lie between 0 and 1.

- $H > 0.55$: Persistent behavior, trending tendency.
- $H < 0.45$: Anti-persistent behavior, mean-reverting tendency.

6.2 Regime Classification

Market regimes emerge from the joint interpretation of the Hurst exponent, market temperature (T), and entropy (S). Thresholds are defined as follows:

$$\boxed{H > 0.55 \quad \& \quad T > T' \quad \Rightarrow \quad \text{Trending Market}}$$

$$\boxed{H < 0.45 \quad \& \quad T < T' \quad \& \quad S < S' \quad \Rightarrow \quad \text{Mean-Reversion Market}}$$

Here, T' divides hot from cold markets, and S' separates ordered from disordered states. In the trending regime, entropy plays a limited role, since persistent directional flow dominates. In the mean-reversion regime, however, entropy must be low (ordered), ensuring that price levels cluster in a structured manner and reversals are reliable. High entropy in a cold market would indicate scattered and illiquid conditions, which weakens mean-reversion signals.

6.3 Trading Frameworks

The choice of technical indicators depends on the regime:

- **Trending Market:** Simple Moving Average (SMA) and Vertex framework. SMA is used to capture directional momentum, while the Vertex framework provides structured entry and exit points.
- **Mean-Reversion Market:** Commodity Channel Index (CCI) and Momentum Oscillator. CCI detects deviations from the average price, while the Momentum Oscillator signals overbought or oversold conditions.

7. Riddles as Market Metaphors

In order to interpret complex dynamics in trading systems, we present three riddles. Each riddle symbolizes hidden structural forces, abstract measures, or transitional behaviors in markets. They serve as metaphors linking the language of **pairs trading**, **thermodynamic regimes**, and **regime-switching frameworks**.

7.1 Riddle 1

"Numbers hum where no eye can see...
Threads twist in ways the mind cannot trace...
Weight never given but multiplies unseen... Silent architect..."

Angles:

- Hidden structural energy, analogous to potential energy in physics, silently governs motion.
- Represents mean-reversion forces that act as the "silent architect" pulling prices together.
- "Weight never given but multiplies unseen" mirrors leverage in liquidity imbalances, producing outsized price moves.
- Symbolizes network effects: even while trading a pair, unseen connections among all assets hum beneath the surface.

7.2 Riddle 2

"I keep two lists: one of breaths, one of sparks...
Read neither as they read themselves...
Fold their tallies through the key I never write, a single notion blinks back..."

Angles:

- "Breaths" = entropy (disorder) and "sparks" = temperature (energy, activity). Their interplay defines market regimes.
- "The key I never write" corresponds to thresholds found via Parameter Generator, which are emergent, not fixed.
- "Single notion blinks back" is the distilled regime signal (trend, mean-reversion, chaos, illiquidity) from noisy order-book data.
- Highlights abstraction: raw numbers hold little meaning until reframed into higher-order constructs like T and S.

7.3 Riddle 3

"Not the rise, nor fall, I claim...
Where the rise begins to wane...
Whisper in the deep..."

Angles:

- Describes **phase transitions**—turning points where one regime weakens into another, captured by adaptive PG frameworks.
- "Whisper in the deep" reflects long-memory in volatility clustering, not just price trends.
- Interprets momentum decay in pairs trading: spreads do not break loudly but soften quietly before mean reversion.
- Links timing of entries/exits with regime logic as well as with subtle structural cues in the market.

8 Backtesting Report Summary

8.1 Strategy 1: Pair Trading (Signal-Based)

Signals:

- $+1 \Rightarrow$ Enter long spread (Buy A, Sell B).
- $-1 \Rightarrow$ Enter short spread (Sell A, Buy B).
- $0 \Rightarrow$ Hold / no trades.

Constraints:

- Trade volume capped by trade logs.
- Net position limited to ± 50 units.
- Discrete adjustments only on signal change.

8.2 Strategy 2: Thermodynamic + Hurst (Volume-Based)

Signals:

- Signal represents target net position.
- Example: Current $+10$, Signal $-2 \Rightarrow$ Sell 12 units.

Constraints:

- Full rebalancing to match target.
- Trade volume capped by trade logs.
- Net position limited to ± 50 units.

Honourable Mention

PLS–Factor Strategy (Fama–French inspired)

Although not used in the final submission, this idea delivered solid paper returns and was novel in our lineup. We framed microstructure signals as *factors* and forecasted the next-tick return from a compact, de-correlated set of components.

Features & Dimensionality Reduction. We used order-book/microstructure features such as order-book imbalance (OBI), bid–ask spread, realized volatility, and depth/pressure. Features were standardized and passed through *Partial Least Squares (PLS)* to extract $K = 4$ latent components $z_{1,t}, \dots, z_{4,t}$ that mitigate multicollinearity while preserving covariance with returns. A simple OBI definition is

$$\mathrm{OBI}_t = \frac{\mathrm{bid\_sz1}_t - \mathrm{ask\_sz1}_t}{\mathrm{bid\_sz1}_t + \mathrm{ask\_sz1}_t}.$$

Linear Factor Model (estimated by OLS). We fit a rolling linear model on the PLS components:

$$\widehat{r}_{t+1} = \alpha_t + \sum_{i=1}^{4} \beta_{i,t}\, z_{i,t},$$

where α_t (intercept) and $\beta_{i,t}$ (factor loadings) are obtained via Ordinary Least Squares.

Trading Rule. At each tick, compute $\widehat{r}_{t+1}$ and trade the sign:

- If $\widehat{r}_{t+1} > 0 \rightarrow$ **Long**; if $\widehat{r}_{t+1} < 0 \rightarrow$ **Short**; otherwise **Hold**.

Probability Gate (concept). We also maintain a short rolling hit-rate of recent forecasts and only trade when this confidence exceeds a high threshold (e.g., 0.8). This acts as a simple regime filter without changing the underlying model.

Why did we not use it? The strategy did not include a position sizing method and if we could optimize the signal generation logic and improve the confidence level at which it was generating a particular signal, this may have significance use in a high frequency trading strategy.